Kinoshi-Tiny Key 25 MIDI Implementation

Version 1.1.0 (Oct. 29 2026)

See also MIDI_Implementation_Chart.md.

- 1. Transmitted Data
- 2. Recognized Receive Data
- 3. Setting Data Format
- 4. Setting Sequence

[H]: Hex, [D]: Decimal

1. Transmitted Data

1-1 Channel Messages

Status [H]	Second [H] ([D])	Third [H] ([D])	Description	Transmitted by	Note
8n	kk (kk)	00 (00)	Note Off	Keyboard	*1
9n	kk (kk)	vv (vv)	Note On vv=1~127	Keyboard	*1, *2
Bn	cc (cc)	vv (vv)	Control Change	Mod Button	*3
Bn	cc (cc)	vv (vv)	Control Change	Pitch+/- Button	*3, *4
Bn	cc (cc)	vv (vv)	Control Change	Sustain Button	*3
Bn	79 (121)	00 (00)	Reset All Controllers	TRANSFER START	*5
Bn	7B (123)	00 (00)	All Notes Off	TRANSFER START	*5
En	bb (bb)	bb (bb)	Pitch Bend Change	Pitch+/- Button	*4

- **n**: MIDI Channel = 0 ~ F
  - USB MIDI and TRS MIDI each transmit on their own channel (TABLE 1, MSB 40/41).
  - Every channel message is transmitted on both USB MIDI and TRS MIDI.
- **vv**: Value

***1**: kk = 00~7F : 0~127

- kk = (Octave + 1) * 12 + Key
  - Octave = -1~9 (selected by OCT-/OCT+ buttons, 4 at power on)
  - Key = 0~24 (0 = leftmost key)
- At power on the keyboard transmits kk = 3C~54 (60~84).
- Keys whose kk would exceed 7F (127) transmit nothing.
- Note Off is transmitted with the same kk as the Note On, even if the octave has been changed while the key is held.

***2**: vv = Keyboard Velocity (TABLE 1, MSB 48 LSB 01).

***3**: cc = CC Number (TABLE 1, LSB 12).

- Mode = Momentary: vv = On Value when pressed, Off Value when released.
- Mode = Toggle: vv = On Value and Off Value alternately on each press.
    - Nothing is transmitted on release.
    - The first press after power on transmits On Value.

***4**:

- Function = Pitch Bend: Pitch Bend Change is transmitted.
    - While pressed, the value changes gradually and is transmitted repeatedly. It takes about 250 ms from center to the end.
    - Pitch+: toward bb,bb = 7F,7F (+8191), then holds.
    - Pitch-: toward bb,bb = 00,00 (-8192), then holds.
    - When released, bb,bb = 00,40 (0) is transmitted immediately.
    - If both buttons are pressed, the last pressed one is effective. Releasing either one returns the value to center.
- Function = CC: Control Change is transmitted (*3).

***5**: Transmitted when TRANSFER START (TABLE 2) is received, in the order All Notes Off, Reset All Controllers. They are transmitted on the channels in effect before the transfer. At the same time the device forgets held keys (no Note Off is transmitted when they are released), resets toggle states and stops Pitch Bend at center. A held key or button takes effect again when pressed again.

Each time a parameter of TABLE 1 is received (2-2), even if its value is unchanged, the following happens:

- Toggle states of all buttons are reset without transmitting anything (the next press transmits On Value).
- If Pitch Bend is not at center or a button is bending, bb,bb = 00,40 (0) is transmitted. The bend stays at center until the button is pressed again.

1-2 Setting Data (NRPN)

Status [H]	Second [H] ([D])	Third [H] ([D])	Description	Transmitted by	Note
B0	63 (99)	mm (mm)	NRPN MSB	Dump	*6
B0	62 (98)	ll (ll)	NRPN LSB	Dump	*6
B0	06 (06)	vv (vv)	Data Entry MSB	Dump	*6
B0	65 (101)	7F (127)	RPN MSB = Null	Dump	*6
B0	64 (100)	7F (127)	RPN LSB = Null	Dump	*6

- Setting data is transmitted on USB MIDI only (not on TRS MIDI).
- Setting data is always transmitted on MIDI Channel 1 (B0), regardless of the USB MIDI / TRS MIDI channel settings.

***6**:

- mm,ll = Parameter (TABLE 1, TABLE 2, TABLE 3)
- vv = Value (TABLE 1, TABLE 2, TABLE 3)

- Transmitted when DUMP REQUEST is received. (2-2, SEQUENCE 2)
- Firmware version (TABLE 3) is transmitted when VERSION REQUEST is received. (SEQUENCE 3)
- ACK is transmitted when SAVE has been completed. (SEQUENCE 1)

2. Recognized Receive Data

- Received on USB MIDI only. TRS MIDI has no receive function.
- Received messages are not echoed (no MIDI Thru).

2-1 Channel Messages

Not applicable (performance messages are not recognized).

2-2 Setting Data (NRPN)

Status [H]	Second [H] ([D])	Third [H] ([D])	Description	Note
B0	63 (99)	mm (mm)	NRPN MSB	*7
B0	62 (98)	ll (ll)	NRPN LSB	*7
B0	06 (06)	vv (vv)	Data Entry MSB	*7
B0	65 (101)	7F (127)	RPN MSB = Null	*8
B0	64 (100)	7F (127)	RPN LSB = Null	*8

Recognized on MIDI Channel 1 (B0) only. NRPN on other channels is ignored.

***7**:

- mm,ll = Parameter (TABLE 1, TABLE 2)
- vv = Value (TABLE 1, TABLE 2)
- The value is applied when Data Entry MSB is received.
- Data Entry LSB (26H), Data Increment (60H) and Data Decrement (61H) are not recognized.
- Undefined parameters and out-of-range values are ignored. No error is transmitted.
- Applied values take effect immediately but are not saved until SAVE is received.

***8**: May be omitted. When RPN MSB or RPN LSB is received (with any value), the NRPN selection is cleared and following Data Entry MSB is ignored until NRPN MSB and LSB are received again.

3. Setting Data Format

One parameter is transferred as the following sequence.

Byte [H]	Description
B0,63,mm	NRPN MSB, mm = Target (TABLE 1)
B0,62,ll	NRPN LSB, ll = Item (TABLE 1)
B0,06,vv	Data Entry MSB, vv = Value (TABLE 1)
B0,65,7F	RPN MSB = Null

Byte [H]	Description
B0,64,7F	RPN LSB = Null

Example: USB MIDI Channel = 10

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B0,63,40, B0,62,00, B0,06,09, B0,65,7F, B0,64,7F
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TABLE 1: Setting Parameter

MSB [H]	LSB [H]	Parameter	Value / Description	Default
40	00	USB MIDI MIDI Channel	00~0F = 1~16	00
41	00	TRS MIDI MIDI Channel	00~0F = 1~16	00
48	01	Keyboard Velocity	01~7F = 1~127	64
50	11	Mod Button Mode	00,01 = Momentary,Toggle	00
50	12	Mod Button CC Number	00~7F = 0~127	01
50	13	Mod Button On Value	00~7F = 0~127	7F
50	14	Mod Button Off Value	00~7F = 0~127	00
51	10	Pitch+ Button Function	00,01 = Pitch Bend,CC	00
51	11	Pitch+ Button Mode	00,01 = Momentary,Toggle	00
51	12	Pitch+ Button CC Number	00~7F = 0~127	00
51	13	Pitch+ Button On Value	00~7F = 0~127	7F
51	14	Pitch+ Button Off Value	00~7F = 0~127	00
52	10	Pitch- Button Function	00,01 = Pitch Bend,CC	00
52	11	Pitch- Button Mode	00,01 = Momentary,Toggle	00
52	12	Pitch- Button CC Number	00~7F = 0~127	00
52	13	Pitch- Button On Value	00~7F = 0~127	7F
52	14	Pitch- Button Off Value	00~7F = 0~127	00
53	11	Sustain Button Mode	00,01 = Momentary,Toggle	00
53	12	Sustain Button CC Number	00~7F = 0~127	40
53	13	Sustain Button On Value	00~7F = 0~127	7F
53	14	Sustain Button Off Value	00~7F = 0~127	00

- Total 21 parameters. Dump is transmitted in the order of this table.
- Mod and Sustain buttons always work as CC (no Function parameter).

- While Function = Pitch Bend, LSB 11~14 are kept and restored when Function = CC.
- Settings are restored to DEFAULT and saved by holding OCT- and OCT+ at power on (factory reset), or when the saved settings are found to be invalid at power on.

TABLE 2: System Command

MSB [H]	LSB [H]	Command	R/T	Value / Description
60	00	DUMP REQUEST	R	00: Transmits all parameters (TABLE 1) and DUMP COMPLETE
60	01	DUMP COMPLETE	T	00: End of dump
60	02	SAVE	R	00: Writes current settings to internal memory
60	03	ACK	T	00: SAVE has been completed
60	04	TRANSFER START	R	00: Editor starts transmitting settings (SEQUENCE 1). Transmits All Notes Off and Reset All Controllers (1-1)
60	05	VERSION REQUEST	R	00: Transmits firmware version (TABLE 3)

- R: Received, T: Transmitted
- Commands with a value other than 00 are ignored.
- Parameters of TABLE 1 are accepted at any time, with or without TRANSFER START.

TABLE 3: Device Information

MSB [H]	LSB [H]	Parameter	R/T	Value / Description
61	00	FIRMWARE VERSION MAJOR	T	00~7F = 0~127
61	01	FIRMWARE VERSION MINOR	T	00~7F = 0~127
61	02	FIRMWARE VERSION PATCH	T	00~7F = 0~127

- The firmware version follows Semantic Versioning (MAJOR.MINOR.PATCH).
- Transmitted in the order of this table. Received values are ignored.

Example: Version 1.1.0

```
B0,63,61, B0,62,00, B0,06,01, B0,65,7F, B0,64,7F,
B0,63,61, B0,62,01, B0,06,01, B0,65,7F, B0,64,7F,
B0,63,61, B0,62,02, B0,06,00, B0,65,7F, B0,64,7F
```

4. Setting Sequence

SEQUENCE 1: Send settings to the device (Editor -> Device)

1. Editor transmits TRANSFER START (60,04,00). The device transmits All Notes Off and Reset All Controllers on USB MIDI and TRS MIDI (1-1, *5).
2. Editor transmits all parameters of TABLE 1 (21 parameters). The device applies each value to the current settings (not saved yet).
3. Editor transmits SAVE (60,02,00). The device writes the current settings to internal memory.
4. After writing is completed, the device transmits ACK (60,03,00). If writing fails, ACK is not transmitted.
5. When ACK is received, the editor reports that the transfer succeeded. If ACK is not received (timeout), the editor shows an error.

SEQUENCE 2: Receive settings from the device (Device -> Editor)

1. Editor transmits DUMP REQUEST (60,00,00).
2. The device transmits all parameters of TABLE 1 (21 parameters) in table order.
3. The device transmits DUMP COMPLETE (60,01,00).
4. When DUMP COMPLETE is received and all 21 parameters are received, the editor applies them. Otherwise (or on timeout) the editor shows an error and keeps the current settings.

SEQUENCE 3: Get the firmware version (Device -> Editor)

1. Editor transmits VERSION REQUEST (60,05,00).
2. The device transmits FIRMWARE VERSION MAJOR, MINOR and PATCH (TABLE 3) in table order.
3. When all three values are received, the editor shows the version. Otherwise (on timeout) the editor shows an error.